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Stock Crash Risk Prediction with Implied Volatility Index: A Comparison of Tree-Based and Transformer-Based Models

Heeseung Chung¹, Cheong Kim^{1, 2}

¹aSSIST University, Seoul 03767, Republic of Korea

²Dong-A University, Busan 49236, Republic of Korea

Corresponding author: Cheong Kim (e-mail: montwins@dau.ac.kr)

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ABSTRACT Stock price crash prediction is important for both risk management and investment decisions. We build machine learning models that predict whether individual stocks will experience a maximum drawdown over 40% within 60 trading days. The dataset covers 2,812 Korean listed stocks from 2015 to 2024. We test nine models: four tree-based ensembles (CatBoost, XGBoost, LGBM, Random Forest) and four deep learning models (CNN, LSTM, GRU, Transformer), plus logistic regression. In 10-fold walk-forward validation, CatBoost shows the best performance (AUC 0.841, KS 0.559, PR-AUC 0.320), while deep learning models lag behind. SHAP analysis indicates that the implied volatility index (VKOSPI) has the strongest predictive power, followed by wavelet features extracted from price data. The main findings are: tree-based models outperform deep learning on this tabular financial dataset; options market volatility helps predict individual stock tail risk; and wavelet transform provides a useful way to combine daily price patterns with quarterly financial data.

INDEX TERMS Market Crash, Individual Stock, Implied Volatility Index, Wavelet Transform, Gradient Boosting Decision Tree

I. INTRODUCTION

According to the Efficient Market Hypothesis [1], market prices reflect all publicly available information. Therefore, predicting future prices using public information is pointless. However, Grossman and Stiglitz [2] brought up the “information paradox.” They argued that complete market efficiency cannot occur due to information costs. Hong, et al. [3] showed that there is a “lag” in which information is slowly reflected in the market. Another factor is human psychology. Barberis, et al. [4] noted that investor psychology often drives markets in irrational ways. For instance, excessive optimism can raise market expectations, while unfounded pessimism can lead to sudden drops [5]. Shleifer and Summers [6] stated that these inefficiencies create investment opportunities to profit. Various studies have shown that markets are efficient only on average, leaving room for market forecasting [7]. This paper examined the prediction of the downside risk for individual stocks. Predicting stock price declines has been a topic of interest for researchers in finance for a long time [8], as it is important not only for risk management but also for investment performance. Regarding the causes of stock price

declines, prior studies have found that negative internal announcements, information asymmetry [9], agency problems [10], supply-demand imbalances, and surprise selling by certain investors are among the causes [11]. Methodologically, statistical models applied for stock price prediction have been widely used. Regression is probably the most frequently used method [12]. But regression analysis does not properly account for the non-linearity and complexity of financial markets. Another technique, based on time-series models of Generalized autoregressive conditional heteroscedasticity (GARCH) [13, 14], was employed. GARCH allows for intrinsic clustering or asymmetry in stock price volatility and identifies market crashes or extreme negative returns [14]. However, these statistical techniques also have their own weaknesses, as they require that the variables follow a certain probability distribution [15]. Machine learning methods are one of the main trends in stock price forecasting [16]. SVM and the decision-tree-based models (Random Forest, Gradient boosting decision tree-based: Models (e.g., XGBoost and LightGBM) work well in learning complex and nonlinear data patterns [17]. In machine

learning, deep learning models have recently become increasingly interesting. Various forms of RNNs [18], such as LSTMs and GRUs, have gained popularity in stock price prediction due to their ability to model long- and short-term dependencies in time series datasets [19]. Transformers and Convolutional Neural Networks (CNNs) are also applied to the discovery of complex local patterns and features in financial time series data [20, 21].

This study contributes to academia in the following ways.

First, although most previous literature has focused on stock price movements, this paper focuses on the mid-term crash risk of individual stocks. To this end, this paper uses a wavelet transform to combine daily stock price records with quarterly financial data for publicly traded companies. In contrast, earlier papers frequently deal only with market or financial data, as Table 1 shows.

Second, this research revisits the usefulness of decision tree-based models in machine learning rather than focusing solely on deep learning, the latest topic. As a vehicle for this, it leverages a range of models, including traditional machine learning models (logistic regression, random forests, XGBoost, LightGBM, CatBoost) and deep learning models (CNNs, LSTMs, GRUs, Transformer-based models). Through statistical validation using the Friedman and Nemenyi tests, this paper statistically demonstrates that boosting-based ensemble models are more useful than deep learning for medium- and long-term stock price crash prediction, especially in handling tabular financial data.

Third, to forecast firm-specific stock price plunges, this paper examines not only financial and stock price information but also the characteristics of the options market. Individual stocks are influenced by and can also affect the overall financial market [22]. They are linked to both how the company is doing overall and how the financial markets as a whole are performing. A strong indicator that shows this relationship is the options market. We shall show in this research that implied volatility data provides significant predictive power regarding the likelihood of a crash in an individual stock, as validated through SHAP feature importance analysis and ablation studies.

II. LITERATURE REVIEW

A. MACHINE LEARNING MODELS FOR STOCK MARKET PREDICTION

Predicting stock prices using machine learning has remained an active research area for a long time. Table 1 provides an overview of machine learning algorithms used to predict stock price direction in the literature. Among these, Random Forest [23], a popular tree ensemble model, is one of the most common techniques. Random Forest is used to control overfitting and to enhance prediction accuracy by assembling several decision trees [24]. Sadorsky [25] used this approach to achieve a high accuracy in predicting the direction of clean energy stocks ETF 20 days ahead. Also, Yin, et al. [26] were

able to forecast long-term directional trends for up to 200 days using an optimized Random Forest method.

Table 1. Stock Prediction Models in Literature

	Machine Learning Model			Deep Learning Model					Features			
	L R	R F	X G B	L G B M	C A T	L S T M	C N N	G R U	T F	S T	O P T	F I N
Sadorsky [25]		✓								✓		
Yin, et al. [26]		✓		✓						✓		
Naik and Mohan [27]			✓									✓
Zhao, et al. [28]			✓	✓						✓		
Jabeur, et al. [29]					✓							✓
Ding and Qin [30]						✓				✓		
Wang, et al. [31]						✓	✓			✓		
Wang, et al. [32]						✓	✓	✓	✓	✓		
Li and Qian [33]						✓		✓	✓	✓		
Hartanto, et al. [34]			✓	✓	✓	✓				✓		
Ji, et al. [35]						✓			✓	✓		
Gezici and Sefer [36]						✓			✓	✓		✓
Kabir, et al. [37]						✓			✓	✓		
Li and Xu [38]								✓	✓	✓	✓	
This paper	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Note: LR = Logistic Regression, RF = Random Forest, XGB = XGBoost, LGBM = LightGBM, CAT = CatBoost, LSTM = Long Short-Term Memory, CNN = Convolutional Neural Network, GRU = Gated Recurrent Unit, TF = Transformer, ST = Stock/Technical, OPT = Options, FIN = Financial.

Gradient Boosting Decision Tree families based on Friedman's theory [39] are presented and discussed: XGBoost, LightGBM, and CatBoost. Such powerful models have also been employed for a variety of financial forecasting problems. Naik and Mohan [27] introduced an XGBoost model to predict whether a stock would fall by more than 10 percent over the next 2-3 days. Zhao, et al. [28] applied a model based on LightGBM to predict the closing price every day, and the model by Jabeur, et al. [29] predicted the financial failure of a listed company after 1 to 3 years. Hartanto, et al. [34] conducted a comparison study of GBDTs and reported

empirical efficiency results for CatBoost and LightGBM in stock price prediction.

Deep learning models are also widely used for stock price prediction. Ding and Qin [30] used LSTM networks to simultaneously predict the next-day daily closing price, daily opening price, high price, and low price. Wang, et al. [31] attempted to predict the closing price the next day with a hybrid model that combines CNN and BiSLSTM, called BiSLSTM(Bidirectional Special Long Short-Term Memory). Recently, interest has shifted toward more complex models based on the Transformer architecture [40]. Wang, et al. [32] proved that the model of the Transformer structure outperformed the traditional RNN family of models. Li and Qian [33] used a GRU-based frequency decomposition neural network and integrated a Transformer model for ultra-short-term prediction over a 5-minute interval. Ji, et al. [35] equipped a generative-decoding-based transformer architecture (Galformer) with the task of predicting stock prices three days ahead, and Gezici and Sefer [36] proposed a transformer model for predicting buy/sell signals. Kabir, et al. [37] used a hybrid deep learning model based on the LSTM and the Transformer to forecast the securities price three days further. Li and Xu [38] adopted GAN model combined with a Transformer-based attention mechanism to predict stock prices in five days.

However, recent research has also criticized deep learning models for their complexity, lack of interpretability, vulnerability to overfitting, and dependence on large amounts of data for training. Salman and Liu [41] theoretically derived the overfitting mechanism in deep learning, and Shwartz-Ziv and Armon [42] argued that deep learning may not be the best choice for tabular data, and tree-like models can learn more robust representations on structured data.

B. FEATURE ENGINEERING

In model building, input features matter as much as fabric does in garment-making. Information from the options market has been used as a predictive factor for stock prices [43]. Pan and Poteshman [44] show that the put-call ratio based on options trading can substantially predict equity returns. Bali and Hovakimian [45] show that the gap between implied and realized volatility can predict individual stock returns. Machine learning models trained on options data can also be used to predict stock prices. For example, Rosillo, et al. [46] predicted S&P 500 market timing utilizing an SVM model applied to the VIX index. In another study, Rasekhschaffe and Jones [47] employed the disparity in implied volatility between put and call options as one of the major features derived from the GBDT model to predict stock prices. However, majority of previous studies were concerned with historical volatility calculated by past data Patel, et al. [48]. Many researchers have focused on predicting volatility rather than using implied volatility to predict stock prices crash [49-52].

Financial data and stock price data have different time intervals. This difference makes it difficult to combine them directly. Forward-filling is one straightforward solution to this problem. The approach here is to extend the last data value from a lower update frequency over the time intervals of a higher update frequency [53]. But such forward filling might introduce spurious staircase patterns in the data. This may result in underestimating or distorting the true volatility of the data [54]. Another popular method is to use dimension reduction, such as PCA, to reduce the stock price time series. J. Cao(2019) used PCA for the compression of stock and financial data for a machine learning model. He determined that this model outperformed traditional portfolio models. Gunduz [55] applied Variational Autoencoders (VAEs) that model the latent space as a probability distribution to forecast the direction of individual stocks. However, these techniques have limitations. PCA cannot capture more complex nonlinear relationships due to its linear nature [56]. Overfitting risk and power consumption are problems with Autoencoders [57]. The wavelet transform is also one of the methods for compressing time-series price data in the stock market. This approach divides the data by frequency to discover patterns at each time scale. Hsieh, et al. [58] proposed the wavelet transform as the input features of an RNN. Likewise, Jothimani, et al. [59] constructed successful stock prediction models using wavelet transform for ANN and SVR models. The wavelet transform is an attractive approach because it can handle nonlinearity and address problems such as overfitting and the computational intensity of other methods, such as autoencoders [60].

C. CONTRIBUTION OF THIS RESEARCH

In recent years, Machine learning research on stock market prediction has focused on deep learning and transformer models. Research that compares traditional and deep learning models to forecast individual stock price mid-term decline risk in the stock market by leveraging financial statement, time-series, and options-market-related information has been relatively scarce. This research responds to the needs of the practice and adds academic value to what has been achieved in existing studies.

III. DATA AND FEATURE ENGINEERING

A. DATA PREPROCESSING

1) DATA SCOPE

The dataset used in this analysis covers 2,812 stocks listed on the Korean stock market from 2015 to 2024. Daily market data and index option market volatility (VKOSPI) for these stocks were collected from the Korea Exchange (KRX), and quarterly financial statements were obtained from the Electronic Disclosure System (DART) operated by the Financial Supervisory Service of Korea. [61].

2) DATA QUALITY AND PREPROCESSING

First, adjusted close price was used to compensate for corporate actions such as dividends and stock splits to avoid distorting the return calculations. Second, stocks with consecutive trading halts for the two quarters (identified by realized volatility, $rvol_{60d}=0$) were excluded from the training data. This filtering removed about 2.1% (2,668) of all observations, preventing abnormal trading conditions from distorting model performance. For VKOSPI, daily closing values were used with no missing values observed during the analysis period. Third, all features were standardized using StandardScaler, fitted only on the training set and applied to validation and test sets to prevent data leakage.

3) DATA PARTITIONING FOR CROSS VALIDATION

In this study, we performed cross-validation using the walk-forward method [62] to evaluate the model's general performance in reflecting the characteristics of time series data. The partitioning was performed using a sliding window method, and each fold was organized to contain temporally contiguous data. We constructed a total of 10 folds by moving a sliding window forward in time, as shown in Fig 1. To address the frequency difference between daily stock prices and quarterly financial data, all observations were extracted based on discrete calendar quarter-ends.

As summarized in Table 2, each fold was set up with a training period of 4 years, a validation period of 6 months, and a test period of 3 months, with a stride between folds of 6 months. To avoid potential information leakage due to the 60-day look-back window used for feature computation, a 3-month buffer period was placed between the validation and test sets, and a 3-month buffer was placed between test periods between different folds to ensure independence. Additionally, to prevent look-ahead bias from financial statement disclosure delays, a 3-month reporting lag was applied to all financial variables, ensuring that the model only uses information publicly available at prediction time.

Walk-forward validation and sliding window techniques are useful in situations where the data order is important, such as in financial time series analysis, and where past data is used to predict the future [62]. These approaches avoid information leakage (look-ahead bias) that can occur when evaluating time series forecasting models. This better reflects the model's true predictive power for future data [63].

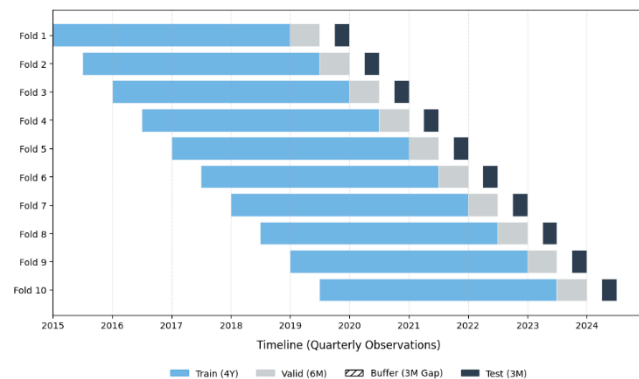


FIGURE 1. 10-fold cross-validation

Table 2. Quarterly Data Spans for Cross-Validation

Fold	Training	Validation	Buffer	Test
1	15.Q1 –18.Q4	19.Q1, Q2	2019.Q3	2019.Q4
2	15.Q3 –19.Q2	19.Q3, Q4	2020.Q1	2020.Q2
3	16.Q1 –19.Q4	20.Q1, Q2	2020.Q3	2020.Q4
4	16.Q3 –20.Q2	20.Q3, Q4	2021.Q1	2021.Q2
5	17.Q1 –20.Q4	21.Q1, Q2	2021.Q3	2021.Q4
6	17.Q3 –21.Q2	21.Q3, Q4	2022.Q1	2022.Q2
7	18.Q1 –21.Q4	22.Q1, Q2	2022.Q3	2022.Q4
8	18.Q3 –22.Q2	22.Q3, Q4	2023.Q1	2023.Q2
9	19.Q1 –22.Q4	23.Q1, Q2	2023.Q3	2023.Q4
10	19.Q3 –23.Q2	23.Q3, Q4	2024.Q1	2024.Q2

B. TARGET VARIABLE DEFINITION

The Target variable in this study is binary. It is defined to take the value of 1 if there is an event that causes the stock price of a given stock to decline by more than 40% within the next 60 trading days at a given point in time, and zero otherwise. In other words, a value of 1 indicates that the risk of a significant stock price decline has been realized in the mid-term. These target variables focus on identifying and predicting the risk of a sharp decline, or “tail risk,” that could result in a substantial loss of asset value for investors [64]

$$MDD = \max_{t \leq \tau \leq \omega \leq t+60} \left(\frac{P_{\tau} - P_{\omega}}{P_{\tau}} \right) \quad (1)$$

where MDD denotes the Maximum Drawdown over the 60 trading days, t denotes the given point in time for the observation, τ and ω are time indices within the 60-trading-day window starting from t , P_{τ} and P_{ω} denote the stock price at time τ , ω , where $\omega \geq \tau$.

This observation period of 60 trading days was set to coincide with the quarterly disclosure cycle of financial statements, the primary learning feature, and the 40% threshold was defined for statistical stability, as shown in Table 3. Preliminary analysis shows that the 20% and 30% thresholds have insufficient discriminatory power to define extreme risk events with an average frequency of occurrence of 37.16% and 16.50%, respectively, while the 50% threshold meets the definition of tail risk with a low frequency of occurrence of 3.17%, the coefficient of variation (CV) of the frequency of occurrence by year was 0.944, which is higher than 40%(0.669).

Table 3. Crash Statistics by MDD Threshold

Threshold (k)	Mean(μ)	Std. Dev. (σ)	CV	Min (yr.)	Max (yr.)
20%	37.16%	8.81%	0.237	25.54%	51.03%
30%	16.50%	7.08%	0.429	10.55%	31.52%
40%	7.01%	4.69%	0.669	3.73%	19.20%
50%	3.17%	2.99%	0.944	1.36%	11.51%

C. FEATURE SET

Stock price plunge prediction models use various characteristics to capture a company's intrinsic value and short-term market trends.

1) WAVELET TRANSFORM

While indicators of a company's financial condition are released quarterly, stock prices fluctuate daily. To overcome this data frequency gap and incorporate the dynamic nature of recent stock price movements into the model, we apply the Wavelet Transform [53, 54] to the daily stock price sequence over the last 60 days. This allows the model to recognize daily stock price patterns even within the quarterly time frame. The stock price time series is decomposed into different frequency bands by the wavelet transform, which is efficient for describing intricacies such as short-term volatility and long-term trend, as shown in Fig. 2. The statistical properties (energy, standard deviation, skewness, kurtosis) of the wavelet coefficients obtained from this transformation of the time series allowed us to represent the essential dynamics of the daily stock prices in a condensed form.

$$cA_{j+1}[k] = \sum_n h[n-2k]cA_j[n] \text{ (Approximation Coefficients)} \quad (2)$$

$$cD_{j+1}[k] = \sum g[n-2k]cA_j[n] \text{ (Detail Coefficients)} \quad (3)$$

Here, where $h[]$ denote a low-pass filter, $g[]$ denotes a high-pass filter, both determined by the chosen mother wavelet, j denotes the level of decomposition, while n and k are indices for the coefficient position, $cA_j[n]$ represents the approximation coefficients at decomposition level j . For the initial level ($j=0$), this corresponds to the original stock price time series. $cD_{j+1}[k]$ are the detail coefficients at the next level, capturing the high-frequency components of the signal, $cA_{j+1}[k]$ are the approximation coefficients at the next level, representing the low-frequency, smoothed components of the signal.

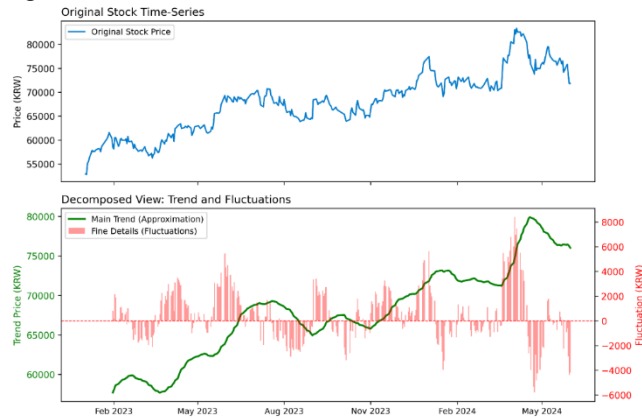


FIGURE 2. 10-fold cross-validation Wavelet Decomposition: Samsung Electronics Stock Example

In this study, we used Daubechies 4 wavelets [65] to perform the decomposition up to the $J=3$ level, resulting in a set of wavelet coefficients $\{cA_3, cD_3, cD_2, cD_1\}$ was obtained. This configuration is widely adopted in financial time series analysis[58, 59]. For each wavelet coefficient array $C = \{c_k | k=1,2,..., M\}$ (where C represents one of cA_3, cD_3, cD_2, cD_1 and M is the number of coefficients in that

array), we compute the following statistical properties Define the mean of the corresponding array C as $\mu_C = \frac{1}{M} \sum_{k=1}^M c_k$.

Energy (E_C): the signal's overall strength in the specified frequency band.

$$E_C = \sum_{k=1}^M c_k^2 \quad (4)$$

where C_k is the k -th coefficient, and M is the total number of coefficients

Standard Deviation (σ_C): reflects how widely the coefficients in that band fluctuate.

$$\sigma_C = \sqrt{\frac{1}{M} \sum_{k=1}^M (c_k - \mu_C)^2} \quad (5)$$

where μ_C is the mean of the coefficients

Skewness (S_C): any asymmetry in the coefficient distribution.

$$S_C = \frac{\frac{1}{M} \sum_{k=1}^M (c_k - \mu_C)^3}{\sigma_C^3} \quad (6)$$

where σ_C is Standard Deviation

Excess Kurtosis (K_C): gauges how peaked the distribution is and how heavy its tails are compared with a normal distribution.

$$K_C = \frac{\frac{1}{M} \sum_{k=1}^M (c_k - \mu_C)^4}{\sigma_C^4} - 3 \quad (7)$$

2) VOLATILITY INDEX

To embed options-market information that can sway individual share prices, we add the Korean volatility gauge VKOSPI to the feature set [66, 67]. VKOSPI is a 30-day implied volatility index calculated from out-of-the-money call and put option prices across a range of strike prices, interpolated between near-month and next-month expirations. The VKOSPI values used in this study are daily closing values provided by Korea Exchange (KRX). No missing values were observed during the analysis period (2015–2024).

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2 \quad (8)$$

$$VKOSPI_{30} = 100 \sqrt{\left(\frac{T_1 \sigma_1^2 (N_2 - 30)}{N_2 - N_1} + \frac{T_2 \sigma_2^2 (30 - N_1)}{N_2 - N_1} \right) * \frac{365}{30}} \quad (9)$$

In this expression, T is the time to expiration in years, r is the continuously compounded risk-free interest rate, and F is the no-arbitrage forward price for the expiration. K_i is the i -th strike price, $Q(K_i)$ is the price of the out-of-the-money (OTM) option for the strike price K_i , σ_1^2 , σ_2^2 are the variances for the near-month and next-month expirations, calculated using the first equation, N_1 and N_2 are the calendar days to expiration for the near-month and next-month options, respectively.

3) FEATURE SET

The resulting stock price-based dynamic characteristics and market volatility indices were combined with static and quasi-static characteristics such as a company's basic financial health (profitability, debt structure, valuation metrics, etc.), supply and demand factors, and industry classification as inputs to the final model. The final list of selected characteristics and their descriptions is shown in Table 4 below.

Table 4. Model Features

Category	Variable Name	Description
Stock Market Data	STCK_PRPR	Daily close price (30% limits)
	PRICE_TO_52W_HIGH	Ratio of the current price to its 52-week high.
	PRICE_TO_52W_LOW	Ratio of the current price to its 52-week low.
	ORGN_NTBY_QTY	Net buy quantity from institutional investors.
	SCRT_GRP_CLS_CODE	industry sector code
	FN_ACML_TR_PBMN	Accumulated trading value.
	Wavelet Features (8-features)	Statistical features (energy, std) from wavelet coefficients of 60-day price data.
Option	VKOSPI	KOSPI Option Volatility Index
Financial & Valuation	ROE	Return on Equity.
	ROE_INC	Change in Return on Equity from the previous period.
	PER	Earnings Yield (per Share).
	PSR	Sales Yield (per Share / Price).
	DR	Debt Ratio.
	DEBT_DEPENDENCY	Debt Dependency Ratio.
	NET_DEBT_RATIO	Net Debt to Market Cap ratio.
	DEBT_COST_RATIO	Debt Cost to Market Cap ratio.
	ICR	Interest Coverage Ratio.
	CSGR	Capital Stock Growth Rate
	TOTAL_ASSET_GROWTH	Total Equity Growth Rate
	MONTH	Calendar month of observation (1-12) for seasonality.

IV. METHODOLOGY

A. MODEL

To identify classifiers that align well with the quirks of financial time series, we reviewed several machine learning families.

1) REGRESSION-BASED MODELS

We begin with the canonical regression classifier, logistic regression (LR). The model forms a linear combination of the explanatory variables and then maps that score through a logistic link, yielding the estimated probability that the target event occurs [68].

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p)}} \quad (10)$$

Here, where X denotes the vector of explanatory variables ($X_1, \dots, X_p$), β_0 is the intercept (or bias) term of the model, $\beta_0, \dots, \beta_p$ denote the regression coefficients (or weights) corresponding to each explanatory variable.

2) TREE-BASED MODELS

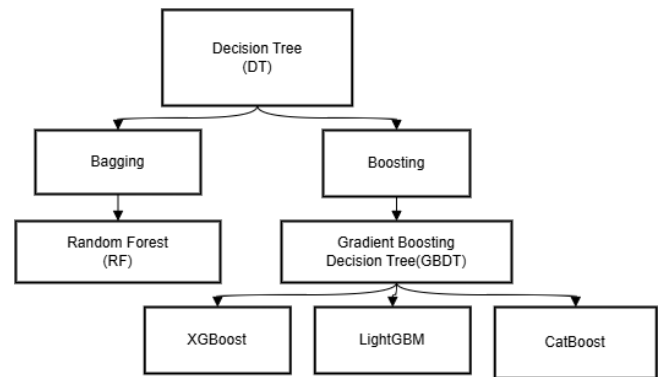


FIGURE 3. Taxonomy of Tree-Based Machine Learning Models

Random Forest (RF), an ensemble model based on Decision Trees (DT) [69], improves prediction accuracy by using bagging across multiple trees [23]. Models like XGBoost [70], and LightGBM [71] are types of Gradient Boosting Decision Trees (GBDT) [39]. The GBDT approach trains multiple trees in sequence, where each new tree tries to correct the error of the previous one, gradually reducing the overall error. CatBoost [72] also presents a variation of the GBDT algorithm. It uses a method called 'Ordered Boosting' which helps to prevent overfitting and enhances the model's generalization performance. Fig. 3 shows the taxonomy of Tree-Based Machine Learning Models.

Feature importance is mostly relevant to tree-based boosting models (XGBoost, LightGBM, CatBoost, etc.), where the tree's splitting structure provides a way to measure how much each variable contributed [73]. Tree-based models train a set of decision trees sequentially to minimize prediction error, and variable importance is based on how often (split), how much information (gain), or how many samples (cover) are involved when that feature is used in splits [70]. Here, this paper computed the importance of the variable by means of the times used (split). However, these methods do not reveal the direction of each feature's impact on predictions. To address this limitation, this paper employed SHAP (SHapley Additive exPlanations)[74], a game-theoretic approach that decomposes each prediction into the contribution of each feature, providing both magnitude and direction of impact.

3) RNN-BASED MODEL

The Recurrent Neural Network (RNN) model proposed by Elman [18] is a versatile deep learning model designed to handle the sequential nature of time-series data. An RNN properly combines information in the previous hidden state and the current input. It is actually described by the following recursive formula: $h_t = f(Wx_t + Uh_{t-1} + b)$. Among RNN architectures, Long Short-Term Memory (LSTM) by Hochreiter and Schmidhuber [75] is outstanding. It includes input, output, and forget gates in the cell itself. This way, the network can decide whether to keep important information and discard what is not necessary. Another popular variant is

the Gated Recurrent Unit (GRU) [76]. The GRU architecture is less redundant, yet it performs similarly to the LSTM.

4) CNN -BASED MODELS

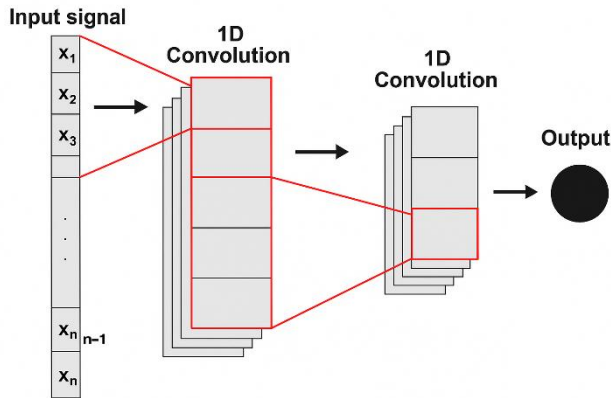


FIGURE 4. One-Dimension Convolution Neural Network Model

Convolutional neural networks (CNNs) [77] are models developed primarily for image recognition. They use specialized filters to automatically and adaptively learn spatial hierarchies of features from input images. Recently, attempts have been made to utilize CNN for time series modeling by converting time series data into image form or directly applying one-dimensional convolutions [78] as shown in Fig. 4.

5) TRANSFORMER MODEL

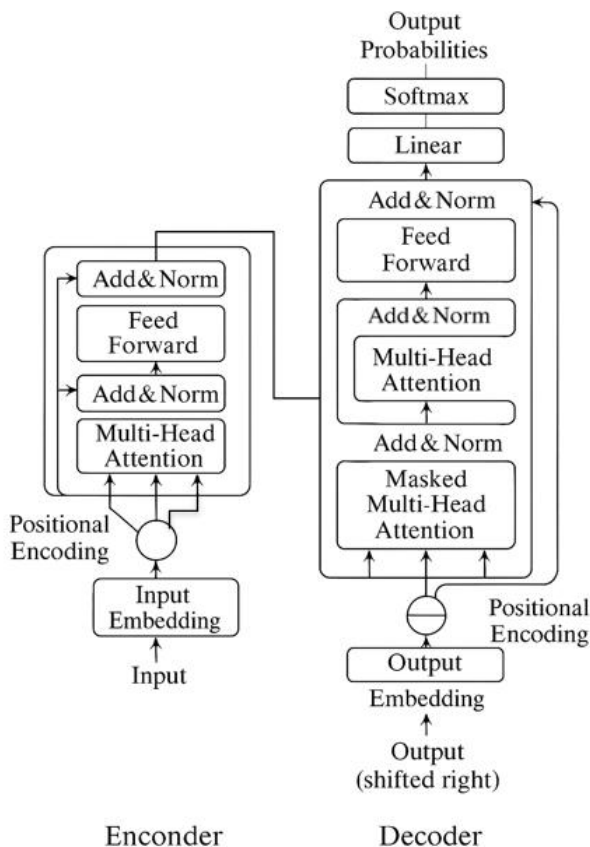


FIGURE 5. Transformer Models

Researchers are now applying Transformers to time-series forecasting as well [79, 80]. Unlike RNNs, a Transformer looks at every element in the sequence at once through self-attention rather than stepping through items one by one [40]. Attention scores are computed with $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V$, where Q , K , and V are the query, key, and value matrices, and d_k is the scaling factor for the dot product.

Fig 5 shows the standard encoder-decoder Transformer architecture. This architecture learns long-range patterns even in very long series and makes good use of parallel hardware. The downside is the time and cost. Transformers demand enormous computational resources. The scaling laws for neural language models [81] clearly illustrate this, demonstrating that computational demands grow immensely with model size. In this study, the Transformer model was adapted for tabular data by incorporating a Feature Embedding layer that treats each summary feature as an independent token, enabling the attention mechanism to learn interactions between features

B. STRATEGY FOR HANDLING CLASS IMBALANCE

The target variable in the training dataset indicates whether the price will fall by more than 40% within the next 60 business days. The distribution of the variable was 17,432 samples in the 'windowdrop=0' (no sharp decline) class and 910 samples in the 'windowdrop=1' (sharp decline occurred) class. The ratio of samples between the two classes was approximately 19.16 to 1. This distribution indicates that the target variable is severely imbalanced.

To curb bias in the target variable class, we applied class weighting during training. By increasing the presence of the minority class, the learner is guided to treat mistakes on crash days as carrying a heavier penalty, thereby improving their ability to spot rare but costly downturns [82].

C. MODEL HYPERPARAMETER

All models were tuned using Optuna [83] with the Tree-structured Parzen Estimator (TPE) sampler. To prevent data leakage, optimization was performed using validation sets from folds 0–6 for training and folds 7–9 for validation, ensuring test sets remained unseen during tuning. For each model, 25–50 trials were conducted with early stopping. Table 5 summarizes the optimal hyperparameters.

Table 5. Hyperparameter Tuning Results

Model	Key Hyperparameters	Optimal Values
CatBoost	learning_rate, depth, l2_leaf_reg	0.015, 10, 26.6
XGBoost	learning_rate, max_depth, subsample	0.026, 4, 0.98
LightGBM	learning_rate, num_leaves, max_depth	0.053, 29, 6
Random Forest	n_estimators, max_depth, min_samples_leaf	382, 15, 9
Logistic Regression	C, penalty	0.27, L1
CNN	learning_rate, dropout_rate, filters	0.0009, 0.43, 16/128
LSTM	learning_rate, dropout_rate, units	0.001, 0.27, 100

GRU	learning_rate, dropout_rate, units	0.002, 0.28, 100
Transformer	learning_rate, num_heads, ff_dim	0.0007, 4, 32

D. EVALUATION METRICS

1) AUC

month and next-month options, respectively. AUC (Area Under the Receiver Operating Characteristic Curve) score [84, 85] ranges from 0.5 for random guessing to 1.0 for perfect separation. Because it is threshold-free, AUC remains a stable yardstick even when the data are unbalanced, making it a sound choice for our setting.

$$AUC = \frac{1}{N_+ N_-} \sum_{i=1}^{N_+} \sum_{j=1}^{N_-} \mathbb{I}(s_i > s_j) \quad (11)$$

Where N_+ is the number of positive (label = 1) instances in the evaluation set, N_- is the number of negative (label = 0) instances in the evaluation set, s_i denotes the prediction score assigned to the i -th positive instance, s_j denotes the prediction score assigned to the j -th negative instance, $\mathbb{I}(s_i > s_j)$ is the indicator function that equals 1 when the score of the positive instance exceeds that of the negative instance ($s_i > s_j$) and 0 otherwise.

2) KOLMOGOROV-SMIRNOV STATISTIC

The KS statistics measure the distance between the cumulative distribution functions of the two classes [86]. Values range from 0 to 1, with higher values indicating greater separation. This measurement is widely used in finance, especially in credit-scoring tasks, and serves as another lens on the model's discriminating power in this research.

$$D = \sup |F_{1(x)} - F_{2(x)}| \quad (12)$$

Here, where $F_{1(x)}$ and $F_{2(x)}$ are the cumulative distributions of the predicted scores for the positive and negative classes.

3) BRIER SCORE

To assess the accuracy of the predicted probabilities, we also report the Brier Score [87]. It is the mean-squared difference between the forecast probability and the actual outcome; smaller values indicate sharper, better-calibrated forecasts.

$$Brier\ Score = \frac{1}{N} \sum_{i=1}^N (f_i - o_i)^2 \quad (13)$$

Here, where f_i is the predicted probability for instance i , o_i is the actual outcome (0 or 1), N is the total number of instances.

4) PR-AUC

For imbalanced datasets, PR-AUC provides a more informative evaluation by focusing on the performance of the minority class.

$$PR-AUC = \int_0^1 P(R) dR \quad (14)$$

where $P(R)$ represents precision as a function of recall.

5) LIFT@10%

Lift measures the concentration of positive instances in the top prediction deciles relative to the base rate. A high lift value indicates that the model is significantly more

effective at identifying potential crashes compared to a random selection.

$$LIFT@K\% = \frac{PRECISION@K\%}{BASE\ RATE} \quad (15)$$

6) RECALL@FPR10%

This metric calculates the recall at a fixed 10% False Positive Rate (FPR), which is highly relevant for risk management applications where the cost of false alarms must be controlled

$$Recall@FPR10\% = R|_{FPR=0.1}$$

where R is the recall achieved when the FPR is constrained to 0.1.

E. STATISTICAL SIGNIFICANCE TESTING

This study gauges whether the observed performance gaps among various machine-learning models are meaningful by applying the Friedman test. The test orders each algorithm's average score and evaluates the null hypothesis that all models share the same average performance (Friedman, 1937; Friedman, 1940).

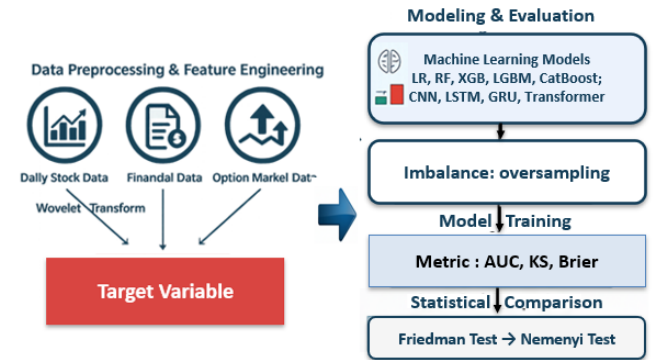


FIGURE 6. MODELING WORK FLOW IN THIS PAPER

When the Friedman statistic rejects the null hypothesis, we turn to a post hoc analysis to identify which pairs of models are truly different. We employ the Nemenyi [88] procedure, a multiple-comparison test that contrasts every classifier with the others. If the gap between the average ranks of two models exceeds the critical difference (CD), their performances are deemed significantly different [89]. Through these statistical tests, the aim is to secure statistical confidence in the observed rank in model performance. Fig. 6 shows the modeling overview of this paper.

V. RESULT

A. MODEL COMPARISON RESULT

Appraising the predictive accuracy of the proposed models, results from each machine learning algorithm are compared to each other using AUC, the Kolmogorov-Smirnov statistic, and the Brier score. Furthermore, PR-AUC, Lift@10%, and Recall@FPR10 are included to ensure a robust evaluation given the significant class imbalance of the data. All experiments were conducted on Google Colab with NVIDIA A100 GPU.

Table 6. Model Evaluation(10-fold Average)

Model	AUC	KS	Brier	PRAUC	Lift	Recall
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CatBoost	0.841	0.559	0.151	0.32	4.74	0.54
XGB	0.839	0.564	0.153	0.311	4.58	0.515
RF	0.834	0.549	0.119	0.321	4.75	0.533
LGBM	0.82	0.525	0.153	0.289	4.41	0.496
LR	0.813	0.516	0.257	0.269	4.4	0.504
CNN	0.805	0.502	0.211	0.259	4.08	0.462
GRU	0.775	0.465	0.2	0.246	3.93	0.426
LSTM	0.758	0.423	0.195	0.244	3.94	0.436
Transformer	0.721	0.36	0.245	0.209	3.2	0.34

As shown in Table 6, CatBoost (AUC = 0.841) achieved the highest AUC, followed by XGBoost (AUC = 0.839), Random Forest (AUC = 0.834), and LightGBM (AUC = 0.820). The four tree-based ensemble models showed comparable performance within a narrow range (AUC 0.82–0.84). In terms of imbalanced data metrics, CatBoost also achieved the highest Lift@10% (4.74) and Recall@FPR10 (0.540), indicating that the top 10% of predictions identified crash events at 4.74 times the base rate, and 54% of actual crashes were detected at a 10% false positive rate.

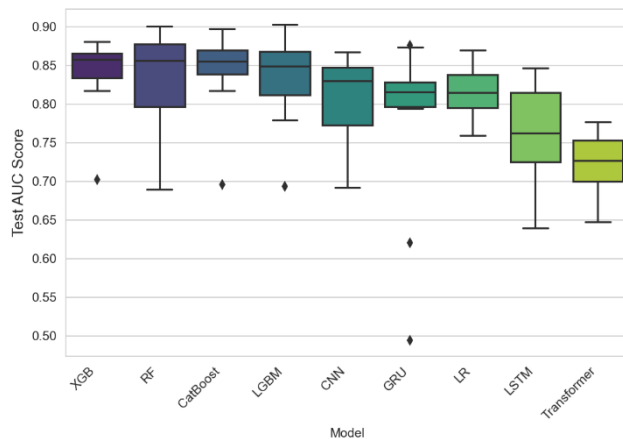


FIGURE 7. Model Performance Comparison on Test AUC (10-Fold Cross-Validation)

Deep learning models performed substantially worse than tree-based models. CNN (AUC = 0.805) showed the best performance among deep models, while GRU (0.775), LSTM (0.758), and Transformer (0.721) lagged behind. The tree-based ensemble models achieved test AUCs above 0.82, whereas deep models ranged from 0.72 to 0.81. This performance gap is also evident in the imbalanced metrics: tree models achieved Lift@10% above 4.4, while deep models ranged from 3.2 to 4.1. These results suggest that for tabular financial data with limited sample size, tree-based ensembles remain more effective than deep learning architectures. Fig 7 shows the overall performance range of the 10-fold evaluation results.

To evaluate statistical significance, we performed a Friedman test based on the test AUC across 10 folds. The test rejected the null hypothesis that all algorithms perform equally ($\chi^2 = 46.75$, $p < 0.0001$), indicating statistically significant differences among models. A Nemenyi post-hoc test at the 0.05 significance level revealed that tree-based ensembles (CatBoost, XGBoost, RF, LightGBM) significantly

outperformed deep learning models (GRU, LSTM, Transformer) and logistic regression. However, there was no statistically significant difference among the four tree-based models, indicating comparable predictive power within this group.

To provide detailed insights into model stability across different market conditions, Table 6-1 presents the fold-wise performance of the best-performing model (CatBoost).

Table 6-1. Fold-wise Performance of CatBoost

Fold	Test Period	Crash Rate	AUC	Lift@10 %	KS
0	2019Q4	0.61	0.696	1.1	0.354
1	2020Q2	0.032	0.817	5.53	0.527
2	2020Q4	0.047	0.84	4.95	0.555
3	2021Q2	0.034	0.874	5.36	0.628
4	2021Q4	0.062	0.85	4.88	0.588
5	2022Q2	0.054	0.838	4.3	0.529
6	2022Q4	0.029	0.864	5.92	0.572
7	2023Q2	0.056	0.861	4.91	0.578
8	2023Q4	0.04	0.897	5.91	0.642
9	2024Q2	0.077	0.871	4.49	0.617

Fold 0 (2019Q4) shows notably lower performance (AUC 0.696) because its prediction target period corresponds to the COVID-19 market crash (2020Q1), resulting in a crash rate of 61% far exceeding the training data average of 6–7%. This extreme distribution shift explains the performance degradation. Excluding Fold 0, the remaining nine folds demonstrate stable performance with AUC ranging from 0.817 to 0.897 across diverse market conditions from 2020 to 2024.

We applied SHAP (SHapley Additive exPlanations) analysis across the four tree-based models (CatBoost, XGBoost, LightGBM, RandomForest) to assess the contribution of each variable. As shown in Fig 8, VKOSPI achieved the highest average absolute SHAP value (0.487), while the wavelet transform features (cd3_energy, ca3_std, cd1_energy) ranked in the top 5, demonstrating their contribution to the crash prediction. While this analysis confirms that the implied volatility of the market as a whole is the most influential predictor, it also captures the significance of the wavelet features. Furthermore, SHAP dependence analysis revealed that VKOSPI maintains a monotonic positive relationship with crash probability, whereas PRICE_TO_52W_LOW exhibits a non-linear relationship

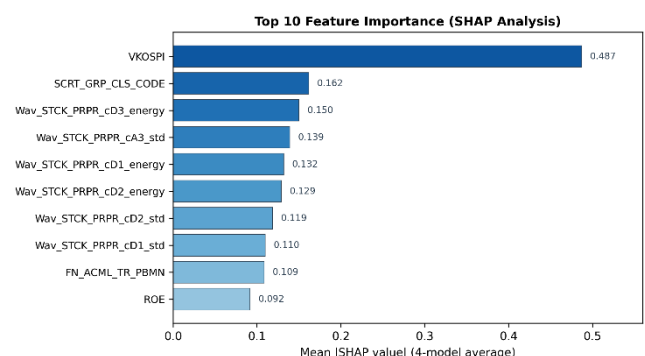


FIGURE 8. SHAP Analysis Result(Average) of 4-tree based Models

B. ABLATION STUDY

In this section, we quantitatively analyze the impact of wavelet features (Wavelet) and the implied volatility index (VKOSPI), which are the main factors contributing to the performance of the proposed models. Ablation analysis was performed by averaging the performance of the four best performing tree-based ensemble models, CatBoost, XGBoost, LightGBM, and Random Forest, and statistical significance was tested by paired t-test with 10-fold cross-validation.

Table 7. Feature Group Performance Comparison

Group	VKOSPI	AUC	vs Base (p-value)	vs Control (p-value)	VKOSPI (p-value)
Base	No	0.7407	-	-	0.068
	Yes	0.8025	-	-	
Base + Control	No	0.7747	0.013	-	0.014
	Yes	0.8336	0.004	-	
Base + Wavelet	No	0.7882	0.01	0.139	0.01
	Yes	0.8365	0.007	0.276	

1) IMPACT OF WAVELET FEATURES:

When wavelet features were combined with traditional financial and market indicators (Base), the AUC improved by about 3.4 percentage points from 0.8025 to 0.8365 in the with VKOSPI condition, which is statistically significant ($p=0.007$). The improvement was also significant in the condition without VKOSPI ($p=0.010$). This indicates that the wavelet transform suppressed the noise in the time series data and effectively extracted the multi-resolution fluctuation information before the stock price plunge.

2) IMPACT OF IMPLIED VOLATILITY (VKOSPI)

The effect of VKOSPI on Base alone was not statistically significant ($p=0.068$), but it was significant when combined with Control or Wavelet features ($p=0.014$ and $p=0.010$, respectively). This suggests that there is an interaction effect between VKOSPI and the price-based features, suggesting that the implied volatility index in the options market captures the tail-risk of individual stocks more effectively when combined with price pattern information.

3) CONTROL VARIABLE COMPARISON:

The Control group, which added simple statistics (60-day return, realized volatility, MDD), also showed significant improvement over the Base ($p=0.013$, 0.004). The direct comparison between Wavelet and Control was not statistically significant ($p=0.139$, 0.276), but Wavelet consistently outperformed (0.7882 vs 0.7747 , 0.8365 vs 0.8336). In addition, individual model analysis without VKOSPI showed marginally significant advantages for CatBoost ($p=0.056$) and Random Forest ($p=0.087$) over the Control group.

C. ROBUSTNESS TEST

To validate the robustness of our findings, we conducted sensitivity analysis across 12 combinations of observation windows (40, 60, 90 days) and crash thresholds (20%, 30%, 40%, 50% MDD). Using CatBoost with the Base+Wavelet+VKOSPI feature set, we performed 10-fold

cross-validation for each configuration. As shown in Table 8, the model achieved test AUC of 0.81–0.90 across all configurations, demonstrating that the predictive performance is robust to different target definitions.

Table 8. Sensitivity Analysis: AUC by Window and Threshold

Window /Threshold	20%	30%	40%	50%
40d	0.836	0.856	0.878	0.902
60d	0.831	0.815	0.837	0.836
90d	0.847	0.812	0.817	0.819

VI. DISCUSSION

In this research, we present a model to help identify downside risk for individual stocks at an early stage. Computationally, ensemble-based tree models performed as well as, or even better than, recent deep learning methods—the addition of learned volatility indices as features proved highly effective at predicting downside risk for individual stocks. The results of this paper have implications for both academic research and application in asset management and risk management.

A. THEORETICAL CONTRIBUTION

Unlike other studies that mainly predict stock price direction or levels, this study focused on predicting stock price tail risk. We explored models that could best predict the risk of a sharp decline in stock prices. This risk is the probability that an individual stock will fall by more than 40% over the next 60 trading days. To combine stock prices and financial data, we used the wavelet transform. This approach mitigates the issues of data distortion and overfitting that arise from forward filling and from autoencoders.

The modeling results showed that tree-based ensemble models, such as catboost, performed better than deep learning models when applied to tabular data that combined stock prices and financial data. The recently popular Transformer model was not effective for the tasks in this study. The stock price time series data we used was wavelet-compressed and integrated with quarterly financial data. It makes it hard for the Transformer model, which excels at modeling temporal patterns in time series data, to showcase its strengths.

A key finding of this study is that the implied volatility index (VKOSPI) from the options market is the most important predictor of individual stock crashes. SHAP analysis revealed that VKOSPI showed the highest contribution among all features. This suggests that overall market sentiment strongly influences tail risk of individual stocks. Although implied volatility is widely known as a leading indicator of market movements [90], few studies have highlighted it as a key factor in predicting individual stock crashes.

Statistical tests supported these findings. Friedman and Nemenyi tests confirmed the superiority of the tree-based model, and paired-samples t-tests demonstrated that wavelet features and VKOSPI contributed significantly to model performance.

B. PRACTICAL CONTRIBUTION

The downside risk prediction model generated by this study also has investment applications. First, the model is a tool for

investment and risk control. It provides numerical scores quantifying the likelihood that each stock will fall sharply. This enables investors to tilt towards high-risk assets and actively manage their portfolios. The model has demonstrated potential in improving the performance of stock indices.

Financial companies can also use it to do a collateral valuation on a stock issue. In some cases, individual stocks are used as collateral for stock-backed loans, which requires assessing the collateral. In such cases, we have to judge the collateral value, apart from the stock's inherent value. Rising stock prices do collateral holders no great favor, while falling stock prices exacerbate the potential for significant impairment of collateral value. Our models can pin down the downside risk of individual stocks and therefore can be a powerful collateral valuation tool, since they target downside risk, not upside potential.

This model can also serve as an early warning system for the risk control department of a financial institution. The risk-management department of a financial institution must have an active plan for when the downside losses in a portfolio reach a certain level. By providing quantitative indicators of individual stock risk, this model can serve as an early-warning system by calculating the ratio of high-downside-risk stocks in the portfolio.

C. ECONOMIC VALUE ASSESSMENT

To assess the economic value of our crash prediction model, we conducted a backtest simulation using a quarterly rebalancing strategy that excludes the top 10% of stocks with the highest predicted crash probability. Transaction costs were assumed at 0.30% one-way (including securities transaction tax of 0.23% and brokerage fee of 0.07%), following Korean market conventions. The results demonstrate that the risk-managed portfolio achieves meaningful risk reduction: crash rate decreased by 1.26 percentage points (from 7.66% to 6.40%), and maximum drawdown (MDD) was reduced by 1.28 percentage points (from 15.36% to 14.08%). Notably, the excluded stocks exhibited substantially higher risk profiles, with a crash rate of 19.01% and MDD of 26.86%, confirming the model's ability to identify high-risk stocks. The Precision@10% of 19.01% represents a lift of 2.48 $\times$ over the base crash rate, indicating effective risk discrimination. However, return improvement was negligible after transaction costs, indicating that the model's primary value lies in risk reduction rather than return enhancement.

Table 9. Economic Value of Risk-Managed Portfolio

Metric	Base	After	Top10%	Improv
Crash Rate	0.0766	0.064	0.1901	+1.26%p
MDD	0.1536	0.1408	0.2686	+1.28%p
Quarterly Return	0.0024	0.0014	0.0068	-0.10%p
Precision@10 %	-	-	0.1901	Lift 2.48x

D. LIMITATION

This study has several limitations. First, due to challenges in obtaining long-term data, the training data contains overlapping periods across folds; however, test sets were independently designed to maintain fold independence. Second, model performance varies with market conditions—during extreme events such as the COVID-19 crash (Fold 3), the exclusion strategy showed reduced effectiveness, suggesting that additional features or regime-specific models may be needed. Third, while the model effectively reduces crash rate and MDD, return improvement after transaction costs was negligible, indicating its primary value lies in risk management rather than alpha generation. Finally, the analysis is limited to the Korean market, and generalizability to other markets requires further validation.

VII. CONCLUSION AND FUTURE WORK

Using Korean stock market data from 2015 to 2024, we compare machine learning models that predict the likelihood of stock prices falling at the end of 60 trading days. The performance of tree-based ensemble models like catboost was higher than that of deep learning-based models, according to walk-forward cross-validation. Furthermore, the implied volatility index, as market data, was more effective than financial features in predicting the risk of a sharp decline 60 trading days hence. The results have implications for both investment performance and risk control.

The performance of the models in this study was assessed using AUC, Brier score and Lift. This is standard practice in machine learning modeling, and the backtest in Section B confirms that the model contributes to actual risk reduction. It is also regrettable that we did not verify the effectiveness of PCA, autoencoders, and the wavelet transform used in this study for stock price data compression. This work shows that deep learning techniques, such as the Transformer, are not always better than machine learning models. However, the hybridization of methods that can generalize (machine learning) and discover nonlinear patterns (deep learning) was not discussed.

Moreover, comparing the performance of several time series compression methods is possible when the characteristic space is widened using autoencoders and other statistical techniques such as PCA. It would also be interesting to evaluate hybrid approaches combining trees and neural networks.

For reproducibility, we have also made the complete source code and experimental sample data publicly available at our GitHub repository [Online].

Available: <https://github.com/hwayobi2020/StockCrash>

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Heeseung Chung received his bachelor's degree in computer science and engineering from Pohang University of Science and Technology in 2007 and completed his MBA in finance at KDI School in 2011. He is currently pursuing a PhD in AI convergence engineering at ASSIST (Seoul School of Integrated Sciences & Technologies). Heeseung Chung has worked at various financial institutions, including securities firms and banks. He has developed derivative product valuation and trading systems, as well as blockchain ecosystems and machine learning-based systems. He has grown into an expert spanning finance and IT. Moving forward, he aims to contribute to the fields of finance and artificial intelligence based on his experience and passion.



Cheong Kim received the B.S. degree in computer education from Sungkyunkwan University, M.S. degrees in business administration from KAIST, and the Ph.D. degree in management information systems from Sungkyunkwan University, Seoul, in 2024. He was a fulltime adjunct professor with aSSIST University from 2024 to 2025. Since September 2025, he has been an Assistant Professor with the Management Information Systems Department, Donga University. His main studies lie in neuroscience-based decision-making, business analytics with an emphasis on deep learning mechanisms, and computational psychology focusing on business decision-making. He has published papers in Information Processing & Management, IT & People, International Journal of Information Management, among others. He is an Academic Editor of the PLOS ONE.

